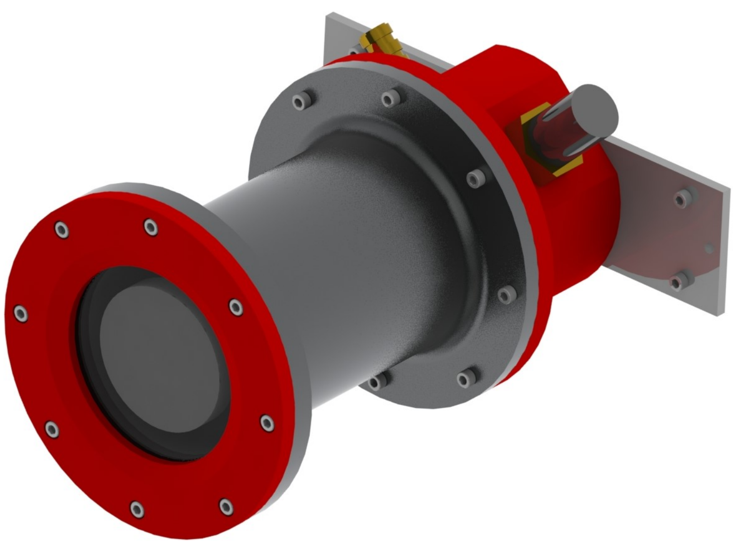
Cornell University Autonomous Underwater Vehicle Team

Spring 2024

Sirius Downcam Enclosure



Technical Report

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Contents

1 Abstract

The downwards-facing camera (known as the ”downcam”) is a particularly useful sensor for the AUV which allows the vehicle to orient and align itself with particular game elements during missions. The enclosure which houses the downcam must contain enough space for the camera, lens, and wiring chosen for the specific design while also maintaining a watertight seal. The enclosure also needs to be mountable to either the UHPV or the frame, and be as small and lightweight as possible while also still being easy to maintenance in potentially time-sensitive situations. Typically, the downcam enclosure consists of five major components: a hull, flange, endcap, camera mounting bracket, and frame mounting bracket (if not directly attached to the UHPV).

For Sirius, an IDS UI-5250CP camera is being housed in the downcam enclosure, which receives power from a HUMG2 SEACON cable and sends data through a MCBH8 SEACON cable. This enclosure is mounted directly to the frame near the torpedoes/droppers assembly and the manipulator so that it can be most useful in assisting these tasks. Internally, the camera sits on a 3D-printed PLA camera mounting bracket, which allows for the most flexibility in lens selection while also being sturdy and reducing vibrations.

2 Design Requirements

2.1 Constraints

- Must completely fit and create a water-tight seal for the UI-5250CP camera, its lens, and all required wired connections
- The viewcone for the UI-5250CP camera must be unobstructed by the enclosure and any surrounding elements within the sub
- Needs to be easily mountable to the Sirius frame
- Must house MCBH8 and HUMG2 SEACON connectors, with flat surfaces to be tightened against

2.2 Objectives

- Be easily machinable on the ”red apron” lathes and mills in the Emerson Manufacturing Lab
- Keep within a max height of approx. 6.5 in to fit cleanly within the Sirius frame and avoid bumping-related damage
- Allow for the UI-5250CP camera to be quickly accessible with minimal disassembly required
- Keep the entire enclosure sturdy and resistent to vibrations, including the internal camera mount

3 Previous Designs

3.1 Killick (2011)

Figure 1: Killick Camera Enclosure

Prior to Killick (2011-2012), Cornell AUVs have featured external camera enclosures in the form of fully-external camera hulls which sealed independently and mounted to the frame of the AUV. This is true for Tachyon (2009-2010) and Drekar (2010-2011). Killick introduced both endcap-mounted and UHPV-mounted camera enclosures for both the forward and downward cameras (known as the ”forecam” and ”downcam” enclosures, respectively). The forecam and downcam enclosures (Figure 1) implemented in Killick’s design are identical and both use O-rings to create face seals to mount to their respective sealing surfaces. The downcam enclosure mounts to the bottom of the UHPV, while the forecam mounts to the endcap of the UHPV. The necessary wiring for each camera is thus able to be contained within the UHPV, allowing for fewer SEACON cables external to the AUV.

3.2 Gemini (2015)

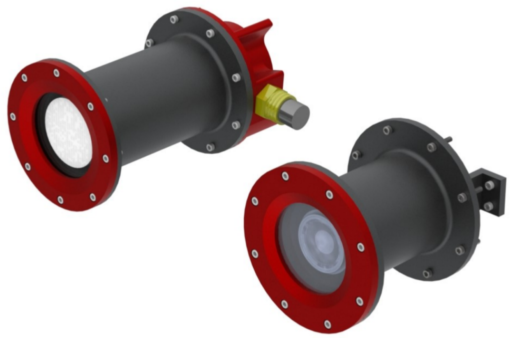


Figure 2: Gemini Camera Enclosures

Gemini's forecam enclosures (Figure 2 on right) used endcap mounted hulls as first used in Killick. The Gemini camera enclosures largely focused on maintaining effective sealing methods and retaining the general model of Killick's designs. Gemini was also the latest AUV that attempted stereoscopic forward vision, and switched to #100 series o-rings for more reliable seals. Gemini reverted back to a fully external downcam enclosure (Figure 2 on left) to account for its UHPV design and the shorter frame. Connectors were also made to extend sideways in Gemini's downcam enclosure design, effectively reducing enclosure height but thickening the rear endcap.

3.3 Castor (2018)

Figure 3: Castor Camera Enclosure

Castor's camera enclosure (Figure 3) took a major detour from previous camera enclosure designs, as it features a large bore seal to secure it directly to the interior of the UHPV. However, Castor was also the first downcam enclosure to contain a bulkhead on its interior camera mounting bracket, designed to keep the camera more stable.

3.4 Odysseus (2019)

Figure 4: Odysseus Camera Enclosures

The forecam (Figure 4 left) and downcam (Figure 4 right) enclosures for Odysseus were similar to their counterparts from Gemini and other intermediate designs. A notable change of the Odysseus camera enclosures was the removal of the internal bulkheads of the camera mounting brackets. The downcam enclosure also featured an updated version of the endcap, which secured the SEACON cables from the side (rather than from the back, as seen with the forecam) and had been elongated to provide more space for wiring. The z-shaped frame mounting brackets also helped to reduce the effective length of the downcam enclosure.

3.5 Leviathan (2020)

Figure 5: Leviathan Camera Enclosures

Leviathan's forecam enclosure (Figure 5 on left) followed eariler designs by sealing its hull directly to the UHPV (rather than an independent endcap). The downcam enclosure (Figure 5 on right) was also quite unique from previous designs; the extruded part of the endcap was elongated and made much smaller in diameter, and the SEACON connectors were angled towards the back once again. Leviathan's downcam also saw the return of the bulkhead.

3.6 Aurora (2021)

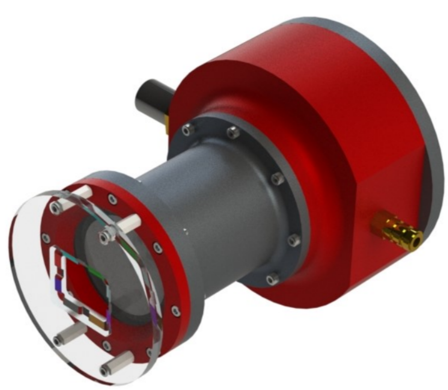


Figure 6: Aurora Downcam Enclosure

The latest camera enclosures first seen on Aurora follow similar designs to their previous counterparts, with exception of the downcam enclosure (Figure 6). Rather than having just an endcap, Aurora's downcam enclosure consists of an "endcap hull" (shown in red) and a thin "endcap" (shown in grey). While this change does greatly increase the outer diameter of the endcap assembly (and its machining time), it also provides more space for the bend radii of the wires and makes wiring the camera much easier. This downcam enclosure also experimented with a "color perception palette" (the outer glass disk with painted grooves), which was used to help adjust the camera to its environment with known color values.

4 High Level Description

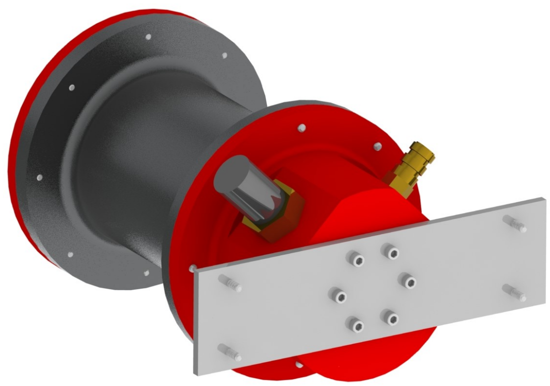


Figure 7: Sirius Downcam

The Sirius downcam enclosure (Figure 7) consists of five major components, including the hull, flange, encap, camera mounting bracket, and frame mounting bracket. The enclosure is sealed through two matching face seals, one between the hull and the flange (with borosilicate glass in between), and the other between the hull and the endcap. Inside of the enclosure, the 3D-printed PLA camera mounting bracket is attached to the interior face of the endcap, and holds the UI camera so that the lens is pushed firmly against the glass to create a large, unobstructed viewcone. The SEACON connectors and frame mounting bracket have been designed to fit cleanly into the rest of the Sirius assembly. This enclosure is notably shorter than most previous camera enclosures (6.325 in), which is mostly due to the space restrictions from the UHPV which sits above the enclosure.

4.1 Hull

Figure 8: Sirius Downcam Hull

The hull (Figure 8) is contructed of 6061-T6 aluminium, and is similar to past camera enclosure hulls. This hull is shorter in length (3 in) than most previous designs, as this is where most of the length reduction of the whole enclosure occurs. The inner diameter of the hull is 2.25 in, which is the minimum amount of space required to comfortably fit the UI camera (This inner diameter is consistent across the endcap and flange as well). The walls of the interior are 0.1 in thick, and the faces on either end of the hull have a diameter of 3.85 in. On both ends of the hull, #037 o-ring grooves are used to create two face seals: one between the hull and the borosilicate glass disk (secured via the flange) and the other between the hull and the endcap. A larger groove (diameter 3 in, depth 0.125 in) is used to make space for the borosilicate glass disk. There are eight #4-40 tapped holes on the side of the hull facing the glass and eight #4 clearance holes on the side of the hull facing the endcap, all of which are through holes. This is so all of the bolts which secure the flange, hull, and endcap together face the same direction, and each can be removed easily while the enclosure is fastened to the frame.

4.2 Flange

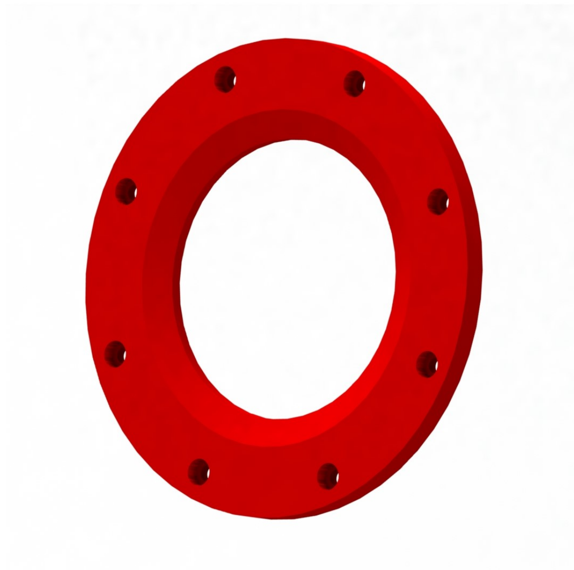


Figure 9: Sirius Downcam Flange

The flange (Figure 9) is made from 6061-T6 aluminium, with an outer diameter of 3.85 in and an inner diameter of 2.25 in (aligning with the hull). The flange is 0.2 in thick, and sports an inner chamfer (this reduces weight and ensures the viewcone is unobstructed, but is mostly for aesthetics). Eight counter-bored #4-40 through holes align with those on the hull, allowing the flange to be well-secured to the rest of the enclosure. The flange applies pressure to the borosilicate glass disk, ensuring the face seal is properly engaged.

4.3 Endcap

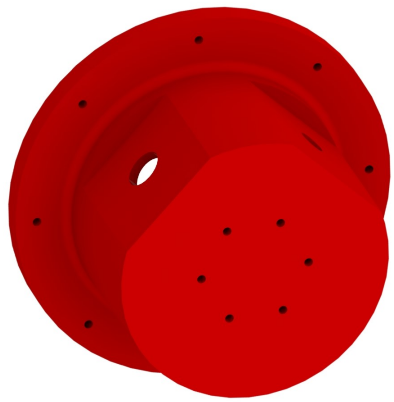


Figure 10: Sirius Downcam Endcap

The endcap (Figure 10) is also made from 6061-T6 aluminium, and is the main part of the enclosure assembly with orients the frame mounting bracket and SEACON cables to ensure the cables do not conflict with the rest of the components on Sirius. The extruded piece of the endcap has an outer diameter of 2.75 in, with two flat surfaces carved to a depth of 0.15 in from this outer diameter. The inner diameter of this component is 2.25 in, which ensures that a minimum thickness of 0.1 in is maintained between the inside and outside of the enclosure. These provide a flat surface for the SEACON connectors to be secured to (to ensure proper sealing), and are angled 90 degrees from one another to avoid conflicting with nearby external components on Sirius. The extruded piece has an internal depth of 1.6 in, which strikes a balance between leaving enough space for wiring and keeping the endcap to a minimal length. On the exterior, eight tapped #4-40 through holes align with those on the hull, allowing for the two parts to be tightly secured, engaging the face seal. There are also six more tapped #4-40 holes (depth 0.25 in) which secure the frame mounting bracket to the endcap. The back of the endcap has a thickness of 0.35 in, ensuring the minimum 0.1 in thickness is maintianed.

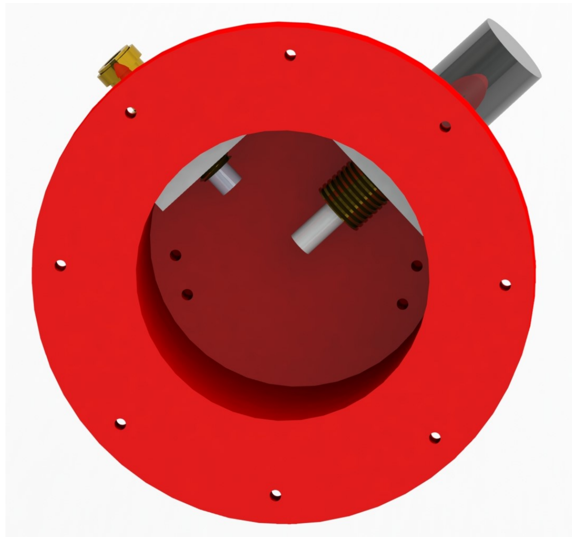


Figure 11: Interior of the Endcap, with the SEACON connectors and supporting 3D-printed PLA surfaces

On the interior of the endcap, there are four more tapped #4-40 holes (depth 0.25 in) which secure the camera mounting bracket to the rest of the enclosure. These holes are arranged so that they are spread out from those on the external face, avoiding any conflict. These interior holes are also arranged so that they have visibility throughout the hull and flange, so the camera mounting bracket could be easily unscrewed without requiring for the entire enclosure to be disassembled. Two additional 3D-printed PLA surfaces (0.25 in thick) are inserted into both SEACON connectors to provide a flat surface for their nuts to be tightened against.

4.4 Camera Mounting Bracket

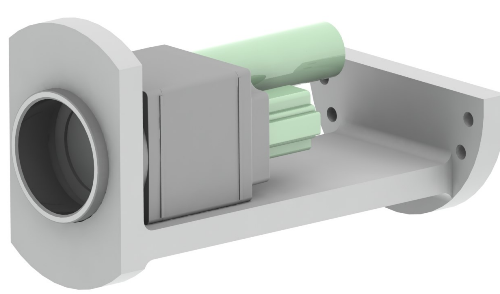


Figure 12: Sirius Downcam Camera Mounting Bracket

The internal camera mounting bracket is the most novel element of the Sirius downcam enclosure. It is 3D-printed from PLA, and resembles an I-beam in its structure. Previous designs for the camera mounting bracket consisted of the ”main bracket” or ”bracket” (usually an L-shaped piece which attaches to the camera and the endcap), and the ”bulkhead” (the slotted cylindrical piece secured around the barrel of the camera, which has often been omitted). This design combines these elements into a single piece, which aims to increase the strength of the bracket and resist vibrations. 3D-printing this camera mounting bracket (rather than machining from aluminium) provides more flexibility to the software and electrical subteams, as it is easy to adjust the current design to fit a different camera or lens and have it be created in a timely manner.

Throughout the camera mounting bracket, a material thickness of 0.25 in is used. The bulkhead is a slotted cylinder with an outer diameter of 2.25 in (matching the inner diameter of the hull) and is 1.4 in wide. The other side of the bracket secured to the endcap is a semicircle (also with a diameter of 2.25 in); this shape helps to give some extra space to the wires while also keeping the bracket sturdy. Connecting these two is a beam (length 3.36 in) with horizontal fillets on either end. Four #4 clearance through holes align with those on the interior of the endcap, and four more M2 clearance through holes on the beam are used to secure the UI-5250CP camera.

4.5 Frame Mounting Bracket

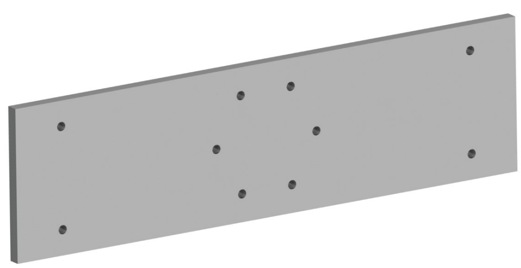


Figure 13: Sirius Downcam Frame Mounting Bracket

The frame mounting bracket is a simple design made from 6061-T6 aluminium. It has a width of 5.5 in, a height of 1.5 in, and is 0.125 in thick. Six central #4 clearance through holes align with those on the endcap, and four outer #4 clearance through holes align with tapped holes on the Sirius frame. The central holes are designed to not conflict with the frame once screws are attached.

5 Current Status

As of early May 2024, all of the parts have been machined or 3D-printed (in PLA), and the 6061-T6 aluminium parts (including the hull, flange, endcap, and frame mounting bracket) have been sent to be anodized. All other integration components, including the screws, o-rings, SEACON connectors, and the borosilicate glass disk, have already been ordered and are currently stored in the ELL. Due to a minor machining error, the endcap has two additional slits removed on its exterior, but this should not effect the performance or the quality of the watertight seal. The hull also needs its fillets to be finished machined, which is planned to be done later this summer. Once the parts return from anodizing, the downcam enclosure will be fully assembled and leak-tested before integrating the electrical components and the camera. After successful leak testing, the electrical and software subteams can ensure that the camera is in full working order before being installed into the Sirius frame and used in full-AUV testing to prepare for competition.

6 Future Improvements

One of the major drawbacks of this camera enclosure design (similar to many previous designs) is that it requires full cylindrical blocks of aluminium stock in order to be machined down to the specifications required. This not only makes the downcam enclosure fairly expensive to make, but also requires extensive machining time to remove all of the excess aluminium for weight savings. Moreover, if a machining mistake is made late in the process (such as an incorrect o-ring groove, which did occur while machining the hull), hours of progress can quickly be lost.

Most of the aluminium tube stock that is offered in the McMaster-Carr or Midwest Steel catalogues is not a suitable alternative to the full cylindrical stock, as the walls of the pipe are not thick enough to be used easily for a face seal (although this may still be possible). Thus, a potential alternative to this design is to use bore seals, which has rarely been seen in past camera enclosure designs (other than the seals secured directly to the UHPV). Currently, another downcam enclosure which makes use of such bore seals is being designed and manufactured for Polaris, so we will be able to test the efficacy of that design. However, given the tight space constraints of Sirius, it still may not have been possible to use a bore seal design for the Sirius downcam, as it does require additional length.

In addition to the stock considerations above, extending the hull just slightly more lengthwise (adding an additional 0.1 to 0.2 in) and adjusting the camera mounting bracket appropriately would likely have given some valuable space to the wiring in the endcap, which will be very cramped. Many past designs mention that they wish they left more space for wiring, citing that having connections at odd angles (rather than on straight, direct paths) adds to the clutter in the back of the enclosure which can be hard to access. While my specific design may have had limited options for potentially addressing this common issue, future camera enclosures should keep these wiring constraints in mind.

Appendices

A Purchased Components

Component	Qty.
Borosilicate Glass Disc	1
#037 O-Ring	1 pack
#4-40 Socket Head Screw, 0.375 in long	1 pack
#4-40 Socket Head Screw, 0.5 in long	1 pack
5/16 in Belleville Washer	1 pack
7/16 in Belleville Washer	1 pack
5/16-24 Nut (Thin)	1 pack
7/16-20 Nut	1 pack

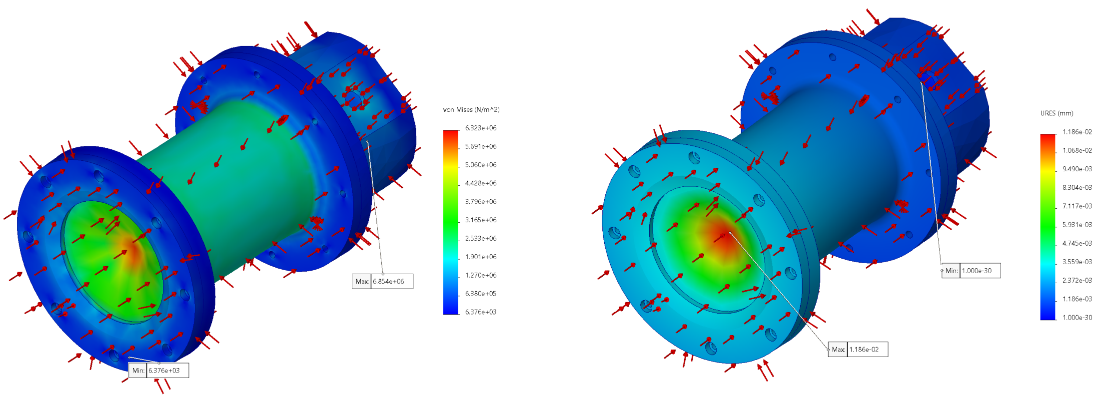
B Finite Element Analysis

B.1 Main Enclosure (Hull, Flange, Endcap)

The main body of the enclosure, including the hull, flange, borosilicate glass, and endcap, was tested with an applied exterior pressure of 30 PSI, which is the approximate force that the enclosure would experience at a depth of 50 feet. Even at a pressure which is well above the maximum that this enclosure is expected to ever experience, the factor of safety is above 40 and the enclosure shows little deformation (less than 1 thou). The maximum stress (994.1 PSI) is also comparable to previous camera enclosures (for reference, the yield strength of aluminium is approximately 35,000 PSI).

Table 1: Main Enclosure Analysis Data

Max Stress	994.1 PSI
Factor of Safety	40.12
Max Deformation	0.000467 in



(a) Equivalent Von-Mises Stress (b) Total Deformation

Figure 1: Main Enclosure SolidWorks Simulation Results

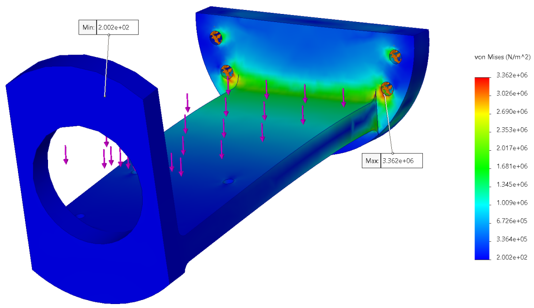
B.2 Camera Mounting Bracket

The internal camera mounting bracket was also analyzed through static force and vibrational frequency SolidWorks simulations. In the static force study, a 10 N force (approximately 4 times the weight of the UI camera) was applied downwards along the face that the camera is mounted to. In this scenario (which could potentially occur during transport), the maximum stress (511.8 PSI) and deformation (0.003309 in) are still well within the acceptable limits for the 3D-printed PLA material (which has a yield strength of approx. 8,700 PSI and 6% elongation at break). In addition, this simulation assumed that bulkhead was not making contact with hull, which may further reduce the stress and deformation.

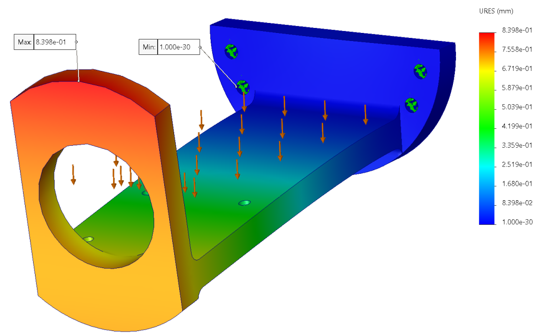
In the vibrational frequency simulation, a probing force of 5 N was applied to the thin exterior faces of the camera mounting bracket. Of the five generated modular shapes which SolidWorks produced, the two more relevant (lowest frequencies) are shown. The mode 1 frequency of 98 Hz could potentially be achieved during transit (such as on a bumpy road), but the addition of the bulkhead (which was also unbound in this simulation) should help to mitigate these resonance issues. The twisting motion of mode 2 is not necessarily restricted by the bulkhead, but the resonant frequency of 272 Hz is unlikely to be achieved in either the transportation or operation of the AUV.

Table 2: Fore Endcap Analysis Data

Max Stress	511.8 PSI
Max Deformation	0.003309 in
Vibrational Frequency (Mode 1)	98.206 Hz
Vibrational Frequency (Mode 2)	272.15 Hz

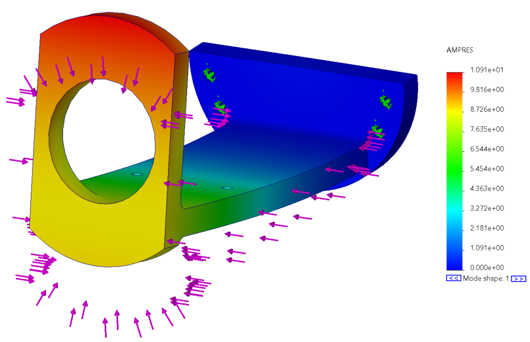


(a) Equivalent Von-Mises Stress

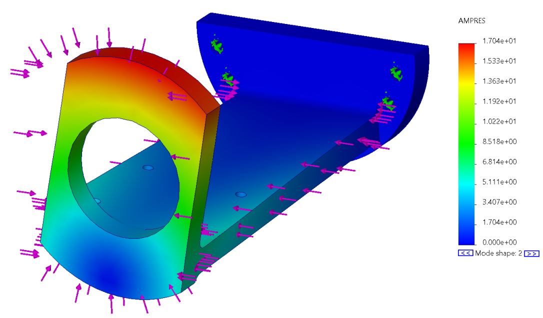


(b) Total Deformation

Figure 2: Camera Mounting Bracket Static Force Simulation Results



(a) Vibrational Frequency Mode 1



(b) Vibrational Frequency Mode 2

Figure 3: Camera Mounting Bracket Frequency Simulation Results